

Parth Mital

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Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

Experience

Concept Marketing

June 2026 – Present

Independent Software Developer (Contract)

Vadara, India

- Rebuilt a legacy ASP.NET WebForms + SQL Server ERP into a **Next.js, TypeScript, Prisma, PostgreSQL** application with **58 API routes**.
- Migrated **114K+ records** from **42 legacy tables** while preserving existing data & application compatibility.
- Built modules for service management, AMC, billing, inventory, reporting, PDF generation, & Excel import/export, with authentication, testing, security, & cloud deployment.
- Delivered a production system used by **60+ daily users**, managing **2.8K customers**, **28K service calls**, & **1.1K AMC contracts**.

LearnLine Edustation (IntelGrader)

June 2026

AI Engineer Intern

Remote

- Built a **YOLOv8** crossout detection system by annotating **100** answer sheets (**277** bounding boxes across **5** classes), reducing OCR clutter by **7.11%**.
- Built an Android document scanning app with **Kotlin, Jetpack Compose, CameraX, OpenCV, TensorFlow Lite, & ONNX Runtime**; trained a **MobileNetV3** model on **10K** images, reducing model size by **82.5%** (**12.5 MB** → **2.2 MB**) & achieving **2.55 ms** inference.
- Built book ingestion & question bank pipelines with **PyMuPDF, EasyOCR, AWS Bedrock, & LLMs**, extracting **1,033** questions from a **280**-page book & generating a **216**-question JEE dataset.
- Built lecture generation pipelines with **LLMs, Stable Diffusion, Text-to-Speech, & Text-to-Video** models, & redesigned the **React** student planner interface.

Projects

🐙 NeuroScan AI: Brain Tumour MRI Analysis System

- Conducted a comparative study of **21 deep learning architectures** for brain tumour detection, classification, & segmentation; authored **2 research papers** under review.
- Built a **3-stage AI pipeline** benchmarking **4,705 MRI scans & BraTS 2020**, achieving **99.59% detection**, **98.58% classification**, & **0.7655 Dice**.
- Engineered a medical imaging platform with MRI inference, 3D visualisation, volumetric analysis, PDF reporting, JWT authentication, & REST APIs.

🐙 Muse: Self-Hosted Music Streaming & Discovery Platform

- Built a **Spotify-inspired streaming platform** with a distributed **4-service architecture** using **Next.js, Fastify, PostgreSQL, & FastAPI**.
- Implemented a **Last.fm recommendation engine** with personalised ranking, **25-track queues**, radio, & up to **44 Tidal resolutions** per request.
- Built a backend with job queues, workers, JWT, Docker, & caching for **10K profiles**, **50K recommendations**, & **100K playback sessions**.

Technical Skills

Languages: C, C++, Java, Python, TypeScript, JavaScript

Coursework: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Digital Systems, Microprocessors

Software: Full-Stack, Frontend, Backend, REST APIs, Authentication, Databases

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Model Training, Feature Engineering, ETL

Cloud & DevOps: AWS, Docker, Cloud Deployment, Containerization